

BatchBase – Project Design Notes

1. Product Description

BatchBase will demonstrate its functionality using a mock pharmaceutical product.

The product is a **dry powder inhaler (DPI)** containing:

- 1 API (active pharmaceutical ingredient)
- 2 excipients:
 - EXP 1
 - EXP 2

2. Manufacturing Workflow

The manufacturing process for the mock product will follow this workflow:

1. Material Supply

A purchase order is created for the materials required for the in-process batch (blend batch).

The required materials are:

- API
- EXP 1
- EXP 2

Each material will have its own material/lot number and manufacturing date.

2. Blending Stage

The required materials are added to the blender:

- EXP 1
- API
- EXP 2

The materials are blended to produce the in-process blend batch.

The blend batch will have its own **in-process batch number**.

The start and end times of the blending process will be recorded.

3. Filling Stage

The blended powder is transferred to the filling process.

The filling process will have a recorded start and end time.

4. Assembly Stage

The filled product is assembled into the final inhaler device.

At this stage, the final product is assigned a **finished product batch number**.

The finished product batch will therefore be linked back to the in-process blend batch from which it originated.

The assembly process will have a recorded start and end time.

3. QC Testing

The mock product will contain three main QC tests.

QC Test 1 – In-Process Blend Uniformity

This test is performed on the in-process blend.

- 10 samples are tested
- Each sample is assessed for potency
- Acceptance range: **90–110% potency**
- The individual sample results will be recorded
- The overall test will receive a pass/fail result

The test will be linked to the **in-process batch number**.

Future versions could include additional information such as:

- Equipment/machine used
- Analyst
- Test method

- Individual sample information

QC Test 2 – Finished Product Assay

This test is performed on the finished product.

- One replicate per batch
- The assay result is recorded
- Acceptance range: **95–110%**

The test will be linked to the **finished product batch number**.

Future versions could include additional information such as:

- Equipment/machine used
- Analyst
- Test method
- Individual sample information

QC Test 3 – Emitted Dose

This test is performed on the finished product.

- 20 repls across the batch
- The individual results are recorded
- Individual acceptance range 75 – 125%
- Mean acceptance range: **85–115%**
- RSD must be **less than 5%**

The test will be linked to the **finished product batch number**.

The database should ideally store the individual results as well as the calculated:

- Mean
- RSD
- Pass/fail result

Future versions could include additional information such as:

- Equipment/machine used
- Analyst
- Test method

4. Initial Database Tables

The initial version of BatchBase will use the following tables.

1. Raw Materials

Stores information about the raw materials used in manufacturing.

Potential information:

- Material/lot number
- Material name
- Manufacturing date

The mock project will assume that the materials are externally manufactured, so the manufacturing date can represent the vendor/manufacturer date.

2. Blend

Stores information about the in-process blending stage.

Potential information:

- In-process batch number
- Material(s) used in the batch
- Blend start date/time
- Blend end date/time

Future versions could include:

- Machine/equipment used
- Operators
- Manufacturing events
- Stoppages
- Additional process parameters

3. Fill

Stores information about the filling stage.

Potential information:

- In-process batch number
- Fill start date/time
- Fill end date/time

Future versions could include:

- Machine/equipment used
- Operators
- Manufacturing events
- Stoppages
- Additional process parameters

4. Assembly

Stores information about the final assembly stage.

Potential information:

- In-process batch number
- Finished product batch number
- Assembly start date/time
- Assembly end date/time

Future versions could include component information such as:

- Component batch/lot number
- Component expiry date
- Component type

5. Batch Crossover

This table will link the **in-process batch number** to the **finished product batch number**.

This relationship is important because it allows BatchBase to trace a finished product batch back through the manufacturing process to the original blend batch and raw materials.

6. QC Test 1 – Blend Uniformity

Stores the results of the in-process blend uniformity test.

Potential information:

- In-process batch number
- Test date
- Individual test results
- Mean/result where appropriate
- Pass/fail

Future versions could include:

- Machine/equipment
- Analyst
- Test method

7. QC Test 2 – Finished Product Assay

Stores the results of the finished product assay.

Potential information:

- Finished product batch number
- Test date
- Result
- Pass/fail

Future versions could include:

- Machine/equipment
- Analyst
- Test method

8. QC Test 3 – Finished Product Uniformity

Stores the results of the finished product uniformity test.

Potential information:

- Finished product batch number
- Test date
- Individual results
- Mean
- RSD
- Pass/fail

Future versions could include:

- Machine/equipment
- Analyst
- Test method

5. Future Development

The initial version will deliberately keep the data model relatively simple.

Future versions could introduce additional manufacturing information such as:

- Equipment and machine information
- Operators
- Manufacturing stoppages
- Process parameters
- Component traceability
- Additional QC tests
- More detailed laboratory results
- Investigation/event records
- Additional batch relationships

The aim of V1 is to demonstrate the core concept of **bringing related manufacturing, material and QC information together so that a batch investigation can be performed from one application.**